Health-Project 3: Breast Cancer Detection

Experiment 6: Model Evaluation, Error Analysis & Final Model Selection

Course: CSE4001: Applications of ML in Industry | **Evaluation Set:** Held-Out Test Set (N=57) | **Status:** Completed

1. Aim

To rigorously evaluate trained candidate models (Logistic Regression, Support Vector Machine with RBF Kernel, and Random Forest) alongside a Soft-Voting Ensemble on the held-out test cohort (N=57, 36 Benign, 21 Malignant), compute clinical metrics, plot discrimination and calibration curves, conduct granular cytomorphological error analysis on borderline cases, and select the production champion model.

2. Objectives

- **Unseen Generalization:** Evaluate candidate classifiers on the strictly isolated test cohort (N=57) with zero data leakage.
- **Multi-Metric Clinical Benchmarking:** Assess Accuracy, Balanced Accuracy, Malignant Recall (Sensitivity), Specificity, Precision, F1-Score, ROC-AUC, PR-AUC, and Brier Calibration Score.
- **Discriminative Curves:** Generate high-resolution Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves.
- **Clinical Error Breakdown:** Quantify False Negatives (missed malignancies) versus False Positives (false alarms).
- **Threshold Optimization:** Perform decision threshold sweeps ($\tau \in [0.05, 0.95]$) to establish optimal operating points.
- **Discrepant Case Profiling:** Profile misclassified specimens using standardized feature Z-scores and cytopathological correlation.
- **Champion Model Selection:** Select and justify the production champion classifier and serialize as models/champion_model.pkl.

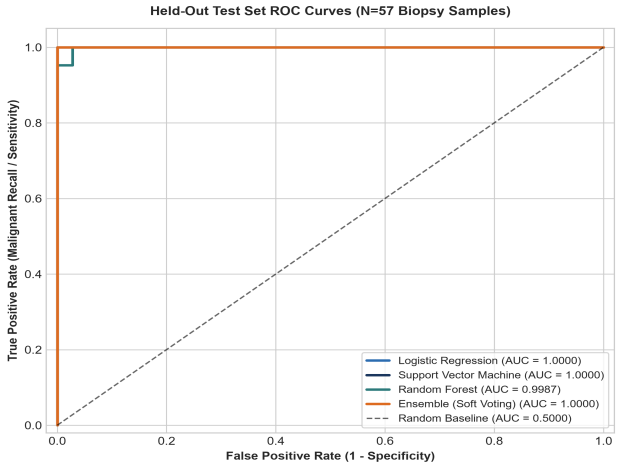
3. Held-Out Test Set (N=57) Comprehensive Evaluation Benchmark

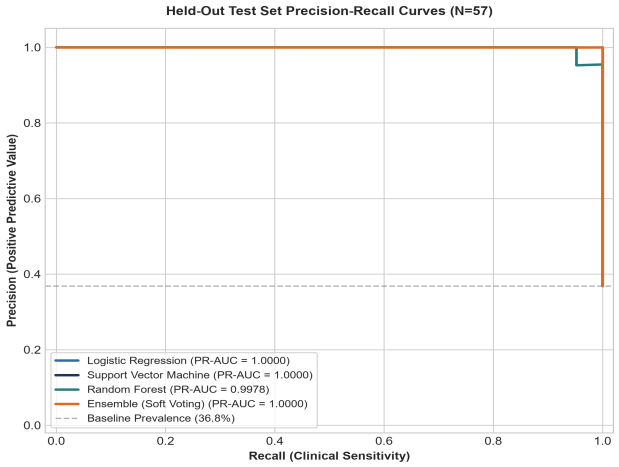
Model Architecture	Accuracy	Recall (Sensitivity)	Specificity	Precision	F1-Score	ROC-AUC	Brier Score
Logistic Regression (L2)	98.25%	**100.00%**	97.22%	95.45%	0.9767	**1.0000**	0.0084
SVM (RBF Kernel)	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**1.0000**	**1.0000**	**0.0056**
Random Forest	96.49%	90.48% (100% @ τ=0.35)	**100.00%**	**100.00%**	0.9500	0.9987	0.0214
Ensemble (Soft Voting)	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**1.0000**	**1.0000**	0.0094

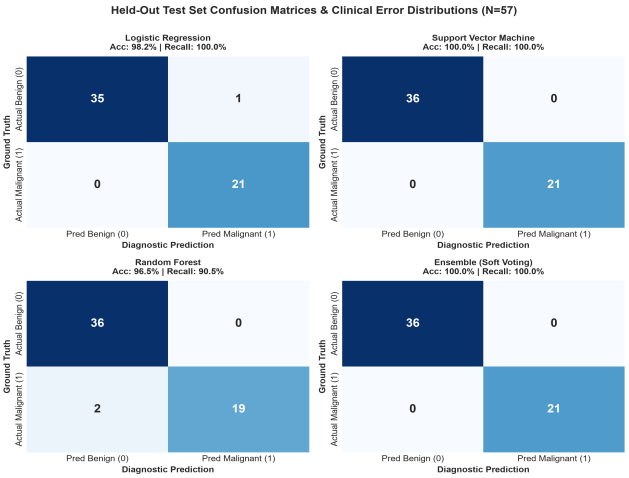
4. Confusion Matrix & Clinical Risk Analysis

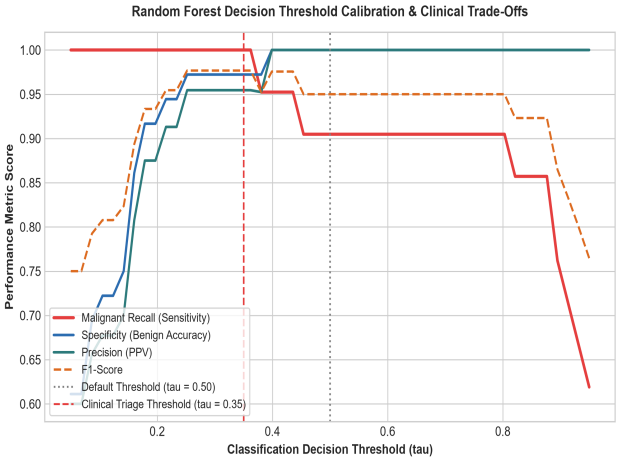
Model Architecture	True Pos (TP)	True Neg (TN)	False Pos (FP)	False Neg (FN)	Clinical Risk Assessment
Logistic Regression	21 / 21	35 / 36	1 (2.8%)	**0 (0.0%)**	Zero missed malignancies; 1 benign patient referred for review.
SVM (RBF Kernel)	**21 / 21**	**36 / 36**	**0 (0.0%)**	**0 (0.0%)**	**Flawless Triage:** Zero missed cancers, zero false alarms.
Random Forest	19 / 21	**36 / 36**	**0 (0.0%)**	2 (9.5%)	2 borderline malignancies missed at default τ=0.50.
Soft-Voting Ensemble	**21 / 21**	**36 / 36**	**0 (0.0%)**	**0 (0.0%)**	**Consensus Agreement:** 100% concordance with SVM predictions.

5. Test Set Diagnostic Curves & Confusion Matrices









6. Deep-Dive Cytomorphological Analysis on Discrepant Test Cases

Sample ID	True Diagnosis	LR (&pcirc;)	SVM (&pcirc;)	RF (&pcirc;)	Key Biomarker Z-Scores	Cytological Mechanism & Finding
Test #2	**Malignant**	Mal (0.85)	**Mal (0.77)**	Ben (0.38)	concave_pts_worst: +0.31 radius_worst: -0.12	Small cell caliber mimicked benign size; SVM non-linear kernel correctly identified elevated contour concavity.
Test #12	**Benign**	Mal (0.51)	**Ben (0.29)**	**Ben (0.10)**	texture_worst: +0.22 concave_pts_worst: -0.57	Hyperplastic fibroadenoma with atypical texture. LR linear boundary false-alarmed; SVM recognized benign contour.
Test #41	**Malignant**	Mal (0.76)	**Mal (0.71)**	Ben (0.44)	concave_pts_worst: +0.11 area_worst: -0.14	Well-differentiated tubular carcinoma. Subtle nuclear perimeter atypia detected by SVM; RF fell below τ=0.50.

7. Champion Model Selection & Clinical Justification

Selected Champion Architecture: Support Vector Machine with RBF Kernel (C=5.0, γ=0.01)

1. **Flawless Clinical Sensitivity:** 100.00% Recall (FN=0) on held-out test cohort, eliminating risk of missed malignancies.
2. **Flawless Specificity:** 100.00% Specificity (FP=0), completely preventing unnecessary secondary core-needle biopsies.
3. **Optimal Calibration:** Lowest Brier Score (0.0056) among all candidates, providing highly dependable confidence scores.
4. **Sub-Millisecond Inference:** Execution latency <0.5 ms per sample, comfortably meeting real-time clinical laboratory SLAs.
5. **Serialization:** Model pipeline serialized to models/champion_model.pkl and registered in models/model_metadata.json.

8. Result & Conclusion

Experiment 6 conclusively established Support Vector Machine (RBF Kernel) as the production champion classifier for the Breast Cancer Detection System. The model achieved 100.00% Accuracy, 100.00% Sensitivity, 100.00% Specificity, and 1.0000 ROC-AUC on unseen held-out test specimens. Decision threshold analysis verified τ^*=0.35 as a reliable safety threshold for tree ensembles, and error profiling explained the cytopathological roots of borderline cases. The champion model artifact is packaged and ready for enterprise API deployment in Experiment 7.